Jigsaws soup: Improved laparoscopic surgical gesture classification, and skill level evaluation using model souping.

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Abstract

Laparoscopy has revolutionised surgery, with faster recovery and less trauma for patients. However, extensive training is needed to gain the required skills, with said training ideally involving simulation-based strategies. A shortage of human expert coaches results in a lack of training opportunities in many jurisdictions. An Artificial Intelligence (AI) coach that can replicate the feedback from a human expert can be utilised to make quality surgical training more accessible. To be effective, this AI coach must classify complex movements into meaningful surgical gestures and then assess the trainee's performance on several parameters. We apply model soup, a recently developed machine learning approach, to improve gesture recognition and surgical score regression from kinematic simulator data. We also apply model soup to improve surgical gesture recognition from video data. By mixing several different 'ingredient' models into a final 'soup' model, performance is increased by 2.84-7.25 percentage points across three surgical tasks over state-of-the-art methods. These improved accuracies bring us closer to a fully autonomous laparoscopic surgery coach, with the potential to dramatically increase the availability of quality training, especially in environments where the availability of teachers and coaches is insufficient.

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